Pure Mathematics Year 2

Chapter 3

Sequences and Series

Arithmetic sequences and series · geometric sequences and series · sigma notation · recurrence relations

Chapter Overview

1. Arithmetic series

Determine the value of

$2 + 4 + 6 + \cdots + 100$.

2. Geometric series

The first term of a geometric sequence is 3 and the second term is 1. Find its sum to infinity.

3. Sigma notation

Determine the value of

$$\sum_{r=1}^{100}(3r + 1).$$

4. Recurrence relations

If $a_1 = k$ and $a_{n+1} = 2a_n - 1$, determine a_3 in terms of k.

Types of sequences

1. $2,\ 5,\ 8,\ 11,\ 14, \ldots$

Name this type of sequence and describe the change between consecutive terms.

1

2. $3,\ 6,\ 12,\ 24,\ 48, \ldots$

Name this type of sequence and describe the change between consecutive terms.

2

3. $1,\ 1,\ 2,\ 3,\ 5,\ 8, \ldots$

How is each new term generated?

3

Definition

An arithmetic sequence has a constant difference between consecutive terms.

Geometric sequences and recurrence relations will be explored later in this chapter.

The fundamentals of sequences

Sequence notation

Consider

$2,\ 5,\ 8,\ 11,\ 14, \ldots$

What does n represent?

What does u_n represent?

For the value 8, state its position and its term notation.

The general term

Let a be the first term and d the common difference.

Moving to position n takes $n-1$ steps.

$u_1 = a,$

$u_2 = a + d,$

$u_3 = a + 2d.$

$$\boxed{u_n = a + (n-1)d}$$

Write down a, d and the relevant position before substituting.

Example 1

$u_n = 55 - 2n.$

(a) Find the first three terms.

(b) Find the first negative term.

1a

1b

Example 2

Find the general term of each sequence.

(a) $6, 20, 34, 48, 62, \ldots$

(b) $101, 94, 87, 80, 73, \ldots$

2a

2b

1. Find the constants

A sequence is generated by $u_n = an + b$, where a and b are constants. Given $u_3 = 5$ and $u_8 = 20$, find a and b.

1

2. Equal differences

For which values of x do -8, x^2 and $17x$ form the first three terms of an arithmetic sequence?

2

Test Your Understanding — Luisa · 7

Edexcel C1 May 2014(R), Q10

Xin has a 14-day training schedule. She runs for A minutes on day 1 and increases her running time by $d + 1$ minutes each day, where A and d are constants.

(a) Show that on day 14 she runs for $A + 13d + 13$ minutes. (2 marks)

a

Yi also has a 14-day schedule. She runs for $A - 13$ minutes on day 1 and increases her running time by $2d - 1$ minutes each day.

(b) Yi and Xin run for the same time on day 14. Find d. (3 marks)

b

Exercise 3A, Y2 pages 61-62 — Luisa · 8

4. The first term of an arithmetic sequence is 14. The fourth term is 32. Find the common difference.

4

5. A sequence is generated by $u_n = pn + q$. Given $u_6 = 9$ and $u_9 = 11$, find p and q.

5

6. An arithmetic sequence has $u_3 = 30$ and $u_9 = 9$. Find its first negative term.

6

7. The 20th term is 14 and the 40th term is -6. Find the 10th term.

7

8. The first three terms are $5p$, 20 and $3p$. Find the 20th term.

8

10. An arithmetic sequence has first term k^2 and common difference k, where $k > 0$. The fifth term is 41. Find k, giving your answer in the form $p + q\sqrt{5}$, where p and q are integers. (4 marks)

10

EXTENSION — Luisa · 11

STEP I 2004, Q5. The positive integers split into five arithmetic progressions:

$A: 1, 6, 11, 16, \ldots$ $B: 2, 7, 12, 17, \ldots$ $C: 3, 8, 13, 18, \ldots$

$D: 4, 9, 14, 19, \ldots$ $E: 5, 10, 15, 20, \ldots$

Give the general term of each progression. Prove that any term in B plus any term in C belongs to E.

Prove that the square of every term in B belongs to D. State and prove a similar claim for the square of every term in C.

Show there are no positive integer solutions:

(i) $x^2 + 5y = 243723$.

(ii) $x^4 + 2y^4 = 26081974$.

i

Series — Luisa · 12

What is a series?

A series is a sum of terms in a sequence. In this syllabus, a series may be finite or infinite.

Arithmetic series

Add terms of an arithmetic sequence:

$2 + 5 + 8 + 11$.

Consecutive terms have a constant difference.

Taylor series

Functions can be expressed as infinite series of powers of x. For example:

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

This is studied in Further Mathematics.

Terminology

A power series is an infinite series of increasing non-negative integer powers of a variable, with constant coefficients. Power series also appear in Further Statistics.

Binomial series

A negative or fractional power can produce an infinite expansion. For $|x| < 1$:

$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} - \cdots$$

This is studied later in Year 2.

Partial sums

The harmonic series is

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$$

A partial sum uses a finite number of terms:

$$H_3 = 1 + \frac{1}{2} + \frac{1}{3} = \frac{11}{6}.$$

Arithmetic Series — Luisa · 13

The sum of an arithmetic series

Let a be the first term, d the common difference and S_n the sum of the first n terms.

Formula

Exam note

Be able to explain this proof. Symmetrically opposite terms add to the same total; the forward-and-reverse argument works for both odd and even n.

See the pairing

Consider $2, 5, 8, 11, 14, 17, \ldots$.
Write the first five terms forwards and backwards.

$S_5 = 2 + 5 + 8 + 11 + 14,$

$S_5 = 14 + 11 + 8 + 5 + 2.$

Add corresponding terms.

Example

Prove the formula

Write the terms forwards and backwards, with $L = a + (n-1)d$. Each corresponding pair has the same sum.

Proof

Alternative Formula — Luisa · 14

Using the first and last terms

Suppose an arithmetic series has n terms, first term a and last term L:

$a + (a + d) + \cdots + L.$

Pair the two copies

Write the terms forwards and backwards. Each pair consists of terms equally far from opposite ends, so each pair sums to $a + L$.

There are n pairs in the two copies together.

Formula

Connection to the earlier formula

The last term is $L = a + (n-1)d$. Substitute this into the first-and-last-term formula.

Connection

Examples — Luisa · 15

Find the sum of the first 30 terms.

Write down a, d and n before substituting. Use S_n to distinguish a sum from the nth term.

1. $2 + 5 + 8 + 11 + 14 + \cdots$

1

2. $100 + 98 + 96 + \cdots$

2

3. $p + 2p + 3p + \cdots$

3

4. Find the minimum number of terms for $4 + 9 + 14 + \cdots$ to exceed 2000.

4

Test Your Understanding — Luisa · 16

Edexcel C1, January 2012, Q9

A company offers two salary schemes for Years 1 to 10 inclusive. Scheme 1 starts at $£P$, increasing by $£2T$ each year. Scheme 2 starts at $£(P + 1800)$, increasing by $£T$ each year.

(a) Show that the ten-year total under Scheme 1 is $£(10P + 90T)$. (2 marks)

a

(b) Both schemes give the same ten-year total. Find T. (4 marks)

b

(c) Scheme 2 pays $£29{,}850$ in Year 10. For this value of T, find P. (3 marks)

c

3. Find the sum of the first 50 even numbers.

3

4. Find the least number of terms for $7 + 12 + 17 + 22 + 27 + \cdots$ to exceed 1000.

4

5. The first term of an arithmetic series is 4. The sum to 20 terms is -15. Find, in any order, the common difference and the 20th term.

Find the common difference

Difference

Find the twentieth term

20th term

6. The sum of the first three terms of an arithmetic series is 12. The twentieth term is -32. Find the first term and common difference.

6

7. Prove that the sum of the first 50 natural numbers is 1275.

7

8. Show that the sum of the first $2n$ natural numbers is $n(2n+1)$.

8

9. Prove that the sum of the first n odd numbers is n^2.

9

10. The fifth term of an arithmetic series is 33 and the tenth term is 68. The sum of the first n terms is 2225.

(a) Show that $7n^2 + 3n - 4450 = 0$.

a

(b) Hence find n.

b

11. An arithmetic series is $(k+1) + (2k+3) + (3k+5) + \cdots + 303$.

(a) Find the number of terms in terms of k.

a

(b) Show that the sum is $\frac{152k+46208}{k+2}$.

b

(c) Given $S_n = 2568$, find k.

c

1. MAT 2008, 1I

For positive integers n, let $S(n)$ be the sum of its digits; for example, $S(723) = 12$. Find

$S(1) + S(2) + \cdots + S(99)$.

1

2. AEA 2010, Q2

An arithmetic series satisfies $S_p = q$ and $S_q = p$, where p, q are positive integers and $p \neq q$.

In terms of p, q, find:

(a) the common difference;

(b) the first term;

(c) S_{p+q}.

Full solution on the next slide.

3. MAT 2007, 1J

For every $n \geq 1$,

$$(n+1) + (n^4 + 2) + \cdots + (n^{10000} + 100) > k.$$

Which follows?

A: $k < 1300$ B: $k^2 < 101$

C: $k \geq 101^{10000}$

D: $k < 5150$

3

AEA 2010, Q2

An arithmetic series has $S_p = q$ and $S_q = p$, where $p, q > 0$ are distinct integers. Find its common difference d, first term a and S_{p+q}.

(a) Common difference

a

(b) First term

b

(c) Sum to $p + q$ terms

c

Recap of Arithmetic vs Geometric Sequences — Luisa · 25

Compare the two sequences

Identify the type of sequence and its constant step.

1. $2, 5, 8, 11, 14, \ldots$

1

2. $3, 6, 12, 24, 48, \ldots$

2

Definition

A geometric sequence is one in which each term is obtained by multiplying the previous term by the same constant ratio.

Quickfire Common Ratio — Luisa · 26

Identify the common ratio r.

Divide a term by the preceding non-zero term.

1. $1, 2, 4, 8, 16, 32, \ldots$

1

2. $27, 18, 12, 8, \ldots$

2

3. $10, 5, 2.5, 1.25, \ldots$

3

4. $5, -5, 5, -5, 5, -5, \ldots$

4

5. $x, -2x^2, 4x^3, \ldots$

5

6. $1, p, p^2, p^3, \ldots$

6

7. $4, -1, 0.25, -0.0625, \ldots$

7

Alternating signs

A negative common ratio makes non-zero terms alternate between positive and negative. For item 5, if $x = 0$, all displayed terms are zero and the ratio cannot be uniquely determined.

Arithmetic sequence

Arithmetic formula

Geometric sequence

Geometric formula

1. Find the tenth and nth terms of $3, 6, 12, 24, \ldots$.

1

2. Find the tenth and nth terms of $40, -20, 10, -5, \ldots$.

2

Tip

Write down a and r before substituting. The exponent is $n - 1$, because there are $n - 1$ multiplications after the first term.

Further Example

Textbook example

The second term of a geometric sequence is 4 and the fourth term is 8. Its common ratio is positive. Find the exact values of the common ratio, the first term and the tenth term.

(a) Common ratio

a

(b) First term

b

(c) Tenth term

c

Tip

Writing $u_2 = 4$ first helps distinguish the second term from the sum of the first two terms.

Further Example — Luisa · 29

The first three terms of a geometric sequence are $3, x, x+6$. All terms are positive.

Find x and the tenth term.

1. Find x

1

2. Find the tenth term

2

Method

Equate consecutive ratios when three consecutive terms are given. Check which solutions satisfy the conditions in the question.

nth term with inequalities — Luisa · 30

Find the first term of the geometric sequence $3, 6, 12, 24, \ldots$ that exceeds one million.

1. Find its position

1

2. Calculate and check the term

2

Tip

The position n must be an integer. Use the inequality to choose the first valid position, then calculate the term.

Test Your Understanding — Luisa · 31

1. A geometric sequence has only positive terms. Its third term is 20 and its fifth term is 80. Find the twentieth term.

1

2. The second, third and fourth terms of a geometric sequence are $x, x+6, 5x-6$.

(a) Find the possible values of x.

(b) Find r, given that it is positive.

(c) Hence find the first term.

2a

2b–c

Exercise 3C , Year 2 Pages 69-70 — Luisa · 32

Exercise 3C, question 3

If $3, x, 9$ are the first three terms of a geometric sequence, find:

(a) the exact value(s) of x;

(b) the exact value(s) of the fourth term.

(a) Second term

a

(b) Fourth term

b

Sign check

No positive-ratio condition is stated. Remember to consider both square roots when solving an equation of the form $x^2 = k$, where $k > 0$.

5. The nth term of a geometric sequence is 2×5^n. Find the first and fifth terms.

5

6. The sixth term of a geometric sequence is 32 and the third term is 4. Find the first term and the common ratio.

6

8. The first three terms of a geometric sequence are $8 - x, 2x, x^2$, where $x > 0$.

(a) Show that $x^3 - 4x^2 = 0$. (2 marks)

a

(b) Find the twentieth term. (3 marks)

b

(c) Is 4096 a term of the sequence? Give a reason. (1 mark)

c

Exercise 3C , Year 2 Pages 69-70 — Luisa · 35

9. A geometric sequence has first term 200 and common ratio $p > 0$. Its sixth term is 40.

(a) Show that $5 \log p + \log 5 = 0$.

(b) Find p to 3 significant figures.

a

11. The first three terms of a geometric sequence are $9, 36, 144$. State, with a reason, whether 383616 is a term of the sequence.

11

Sum of the first n terms of a geometric series — Luisa · 36

Arithmetic series

Arithmetic formula

Geometric series

Geometric formula

Proof of the geometric-series formula

Write the sum, multiply by r, then subtract.

Proof

Examples — Luisa · 37

Find the sum of the first ten terms of each geometric series.

Use $S_n = \dfrac{a(1 - r^n)}{1 - r}$ for $r \neq 1$.

1. $3 + 6 + 12 + 24 + 48 + \cdots$

1

2. $4 + 2 + 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$

2

Check

A term u_n and a sum S_n are different quantities. Identify which one the question asks for before choosing a formula.

Harder Example — Luisa · 38

Find the least value of n such that the sum of $1 + 2 + 4 + 8 + \cdots$ to n terms exceeds $2\,000\,000$.

1. Form and solve an inequality

1

2. Verify the boundary

2

Why logarithms?

Taking logarithms lets us solve for an unknown exponent. Since the base 2 exceeds 1, the inequality keeps its direction.

Test Your Understanding — Luisa · 39

Edexcel C2 June 2011, Q6

The second and third terms of a geometric series are 192 and 144 respectively.

Find (a) the common ratio (2 marks), (b) the first term (2 marks), and (d) the smallest value of n for which the sum of the first n terms exceeds 1000 (4 marks).

a

b

d

Exercise 3D, Pure 2 Pages 72-73 — Luisa · 40

4. The sum of the first three terms of a geometric series is 30.5. If the first term is 8, find the possible values of r.

4

5. Find the least value of n such that $3 + 6 + 12 + 24 + \cdots$, to n terms, exceeds 1.5 million.

5

6. Find the least value of n such that the sum

$5 + 4.5 + 4.05 + \cdots$

to n terms exceeds 45.

Set up the sum

The first term is $a = 5$ and the common ratio is $r = 0.9$.

6

7. A geometric series has first term 25 and common ratio $\frac{3}{5}$. Its sum to k terms is greater than 61.

(a) Show that $k > \frac{\log(0.024)}{\log(0.6)}$.

(b) Find the smallest possible value of k.

7a

7b

Extension — MAT 2010, 1B

The sum of the first $2n$ terms of

$1,\ 1,\ 2,\ \frac{1}{2},\ 4,\ \frac{1}{4},\ 8,\ \frac{1}{8}, \ldots$

is which expression?

A: $2^n + 1 - 2^{1-n}$

B: $2^n + 2^{-n}$

C: $2^{2n} - 2^{3-2n}$

D: $\frac{2^n - 2^{-n}}{3}$

Extension

Divergent vs Convergent

What happens to each running total?

A series converges if its partial sums approach a finite limit. Otherwise it diverges.

1. $1 + 2 + 4 + 8 + 16 + \cdots$

1

2. $1 - 2 + 3 - 4 + 5 - 6 + \cdots$

2

3. $1 + 0.5 + 0.25 + 0.125 + \cdots$

3

Beyond the A level syllabus — for interest

4. The harmonic series:

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$$

4

5. The Basel problem:

$$1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots$$

5

Sum to Infinity

Compare two geometric series

$1 + 0.5 + 0.25 + \cdots$ converges to 2, whereas $1 + 2 + 4 + \cdots$ grows without bound.

A geometric series with non-zero first term converges when

$$\boxed{|r| < 1} \quad \text{or} \quad -1 < r < 1.$$

What happens to r^n as $n \to \infty$ when $|r| < 1$?

Derive the sum to infinity

Start with the finite sum:

$$S_n = \frac{a(1 - r^n)}{1 - r}.$$

Use the limit of r^n to obtain S_∞.

Quickfire Examples

For each geometric series, state the first term a, common ratio r and sum to infinity. Check $|r| < 1$.

1. $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \ldots$

1a

1r

1 sum

2. $27, -9, 3, -1, \ldots$

2a

2r

2 sum

3. $p, p^2, p^3, p^4, \ldots$, where $-1 < p < 1$

3a

3r

3 sum

4. $p, 1, \frac{1}{p}, \frac{1}{p^2}, \ldots$. State when the series converges.

4a

4r

4 sum

Further Examples

1. The fourth term of a geometric series is 1.08 and the seventh term is 0.23328.

(a) Show that the series is convergent.

(b) Find its sum to infinity.

1a

1b

2. A geometric series has first term a, common ratio r, $S_4 = 15$ and $S_\infty = 16$.

(a) Find the possible values of r.

(b) All terms are positive. Find a.

2a

2b

Remember: the exponent is n in the sum formula, but $n - 1$ in $u_n = ar^{n-1}$.

Edexcel C2 May 2011, Q6

The second and third terms of a geometric series are 192 and 144 respectively. Find:

(a) the common ratio (2 marks);

(b) the first term (2 marks);

(c) the sum to infinity (2 marks);

(d) the smallest n for which the sum of the first n terms exceeds 1000 (4 marks).

a

b

c

d

2. A geometric series has first term 10 and sum to infinity 30. Find the common ratio.

2

3. A geometric series has first term -5 and sum to infinity -3. Find the common ratio.

3

4. A geometric series has sum to infinity 60 and common ratio $\frac{2}{3}$. Find its first term.

4

5. A geometric series has common ratio $-\frac{1}{3}$ and sum to infinity 10. Find its first term.

5

6. Find the fraction equal to the recurring decimal $0.\overline{23} = 0.232323\ldots$

Use a geometric series

Write each repeated block as a separate term, then identify a and r.

6

Exercise 3E, Pure Y2 Pages 75-76 — Luisa · 51

1. MAT 2006, 1H

How many solutions does

$2 = \sin x + \sin^2 x + \sin^3 x + \cdots$

have for $0 \leq x < 2\pi$?

1

2. MAT 2003, 1F

Two players take turns to throw a fair six-sided die until one scores a six. Find the probability that the first player is the first to score a six.

2

3. Extension

Determine

$x = 1 + \frac{2}{2} + \frac{3}{4} + \frac{4}{8} + \frac{5}{16} + \cdots$

Hint: multiply, then subtract as in the geometric-sum proof.

3

Sigma Notation — Luisa · 52

Read sigma notation

$$\sum_{r=1}^{5}(2r+1)$$

The symbol

The Greek capital letter Σ means **sum**.

The limits

Start at $r = 1$, finish at $r = 5$, increasing r by 1 each time.

The expression

Evaluate $2r + 1$ for each value of r, then add the results.

Expansion

Recognise the series

A linear expression in the summation variable gives an arithmetic series. Establish its first term a, common difference d and number of terms before applying the sum formula.

Determining the value

Luisa · 53

1. $\sum_{n=1}^{7} 3n$

1

2. $\sum_{k=5}^{15} (10 - 2k)$

2

3. $\sum_{k=1}^{12} 5 \times 3^{k-1}$

3

4. $\sum_{k=5}^{12} 5 \times 3^{k-1}$

Either start with the $k = 5$ term, or subtract the first four terms from the first twelve.

4 sum

Testing Your Understanding

Luisa · 54

Solomon Paper A

Evaluate

$$\sum_{r=10}^{30} (7 + 2r).$$

(4 marks)

Method 1: Direct

Identify the first term, last term and number of terms.

Method 1

Method 2: Subtraction

Sum from 1 to 30, then subtract the terms from 1 to 9.

Method 2

On your calculator

Use the summation function

The ClassWiz and Casio Silver calculators have a Σ function. On the CG50, use **MATH**, then the next-menu arrow, then $\Sigma($.

Enter the expression, summation variable, lower limit and upper limit in the calculator's summation template.

Try it

$$\sum_{k=5}^{12} 2 \times 3^k$$

Expression: 2×3^k

Variable: k

Lower limit: 5

Upper limit: 12

Calculator result

Check the result

The first term is $2 \times 3^5 = 486$, the common ratio is 3, and there are $12 - 5 + 1 = 8$ terms.

Formula check

Exercise 3F, Pure 2, Pages 77-78

6. Show that

$$\sum_{r=1}^{n} 2r = n + n^2.$$

6

7. Show that

$$\sum_{r=1}^{n} 2r - \sum_{r=1}^{n} (2r - 1) = n.$$

7

Exercise 3F, Pure 2, Pages 77-78 — Luisa · 57

9. Find

$$\sum_{r=20}^{\infty} 200\left(\frac{1}{4}\right)^r.$$

9

10. Given

$$\sum_{r=1}^{k}(8+3r)=377,$$

(a) show that $(3k+58)(k-13)=0$;

(b) hence find k.

10a

10b

Recurrence Relations — Luisa · 58

Two ways to define a sequence

Position-to-term rule

$u_n = 2n^2 + 3$

Substitute the position n to find any term directly.

Position-to-term

Term-to-term rule

$u_{n+1} = 2u_n + 4, \qquad u_1 = 3$

Double the previous term, then add 4. This is a **recurrence relation**.

Recurrence

An initial value is essential.

A recurrence describes how to move between terms. The starting value determines which sequence it generates.

Example — Luisa · 59

Edexcel C1 May 2013 (R), Q6

$x_1 = 1, \qquad x_{n+1} = x_n^2 - kx_n \quad (n \geq 1).$

Here k is constant. (a) Find x_2. (b) Show that $x_3 = 1 - 3k + 2k^2$.

Given $x_3 = 1$, (c) find k; (d) find $\sum_{n=1}^{100} x_n$.

a–b

c

d

Recurrences need not be arithmetic or geometric. Look for a repeating pattern before choosing a sum formula.

Test Your Understanding — Luisa · 60

Edexcel C1 January 2012, Q4

$x_1 = 1, \qquad x_{n+1} = ax_n + 5 \quad (n \geq 1),$

where a is constant.

(a) Write down x_2. (b) Show that $x_3 = a^2 + 5a + 5$.

Given $x_3 = 41$, (c) find the possible values of a.

a–b

c

1. AEA 2011, Q3

$u_1 = k$, $u_{2n} = pu_{2n-1}$, $u_{2n+1} = qu_{2n}$, for $n \geq 1$.

(a) Write the first six terms.

(b) Show that, for $pq \neq 1$,

$$\sum_{r=1}^{2n} u_r = \frac{k(1+p)(1-(pq)^n)}{1-pq}.$$

(c) With $[x]$ the integer part of x, find

$$\sum_{r=1}^{\infty} 6\left(\frac{4}{3}\right)^{[r/2]}\left(\frac{3}{5}\right)^{[(r-1)/2]}.$$

a

2. MAT 2014, 1H

$F(1) = 0$. For each $n \geq 2$, add to $F(n-1)$:

2 if $2 \mid n$ but $3 \nmid n$;

3 if $3 \mid n$ but $2 \nmid n$;

4 if both divide n;

0 if neither divides n.

Find $F(6000)$.

2

3. MAT 2016, 1G

$$x_0 = 1, \qquad x_n = \sum_{k=0}^{n-1} x_k$$

for $n \geq 1$. Determine

$$\sum_{k=0}^{\infty} \frac{1}{x_k}.$$

Work out the first few terms.

Full worked solution on the next slide.

MAT 2016: Sum of previous terms

$$x_0 = 1, \qquad x_n = \sum_{k=0}^{n-1} x_k \quad (n \geq 1).$$

Find $\displaystyle\sum_{k=0}^{\infty} \frac{1}{x_k}$.

Pattern

Sum

Increasing, decreasing and periodic sequences — Luisa · 63

Strictly increasing

$u_{n+1} > u_n$ for every n.

Strictly decreasing

$u_{n+1} < u_n$ for every n.

Periodic

$u_{n+k} = u_n$. The least positive such k is the order.

State whether each sequence is increasing, decreasing or periodic. Give the order if periodic.

(a)

$u_{n+1} = u_n + 3, \quad u_1 = 7.$

a

(b)

$u_{n+1} = u_n^2, \quad u_1 = \frac{1}{2}.$

b

(c)

$u_{n+1} = \sin(90n^\circ).$

c

Terminology: "strictly" excludes equal consecutive terms. A constant sequence is neither strictly increasing nor strictly decreasing.

Exercise 3H, Pure 2, Pages 82-83 — Luisa · 64

Exercise 3H, Q3

$u_{n+1} = ku_n, \qquad u_1 = 5.$

Find the range of values of k for which the sequence is strictly decreasing.

Answer

Why no other values

Exercise 3H, Q5

$a_n = \cos(90n^\circ), \qquad n \geq 1.$

(a) Find the order of the sequence.

(b) Find $\sum_{r=1}^{444} a_r$.

a

b

Modelling **Luisa · 66**

Bruce models his first-year profit as £20,000, rising by £5,000 each year until it reaches £100,000, then remaining constant.

(a) Find his total profit over 20 years. (b) Give a limitation of this model.

(c) Instead, model profit as increasing by 5% each year. Find the 20-year total.

a

b

c

Geometric Modelling Example

Luisa · 67

A sheet of paper is initially 0.5 mm thick. Each fold doubles its thickness.

(a) Find its thickness after 4 folds.

(b) Find its thickness after 20 folds.

(c) Comment on whether this is realistic for a sheet of A4 paper.

a

b

c

Test Your Understanding

Luisa · 68

Edexcel C2 January 2013, Q3

A company predicts profit of £120,000 in 2013, rising by 5% each year.

(a) Show that the predicted 2016 profit is £138,915.

(b) Find the first year with yearly profit exceeding £200,000.

(c) Find total predicted profit from 2013 to 2023 inclusive, to the nearest pound.

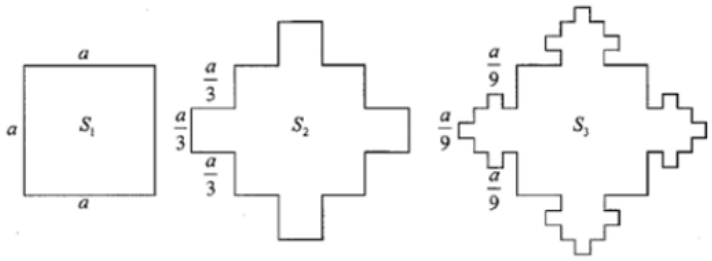
a

b

c

AEA 2007, Q5: Model snowflakes

S_1 is a square of side a. For S_2, replace the middle third of each edge with a new square of side $a/3$. At each later stage, repeat this on each external edge of the squares added at the previous stage, using squares one-third as large.



(a) Show that S_4 adds 36 squares of side $a/27$.

(b) Show that $P_2 = 20a/3$ and $P_3 = 28a/3$.

(c) Find the perimeter P_n.

a

b

c